

Residual Spectral Graph Learning for Connectivity-Loss Estimation under Multi-Edge Disruptions

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Problem and research question

When several road links fail together, how much global connectivity is lost? Exact recomputation is reliable but costly across many scenarios. A first-order spectral approximation is fast and interpretable, but ignores nonlinear interactions. We test whether a GNN should predict loss directly or learn only a correction to the mathematical prior.

Mathematical target

For graph G and failed edge set S , algebraic connectivity is the second Laplacian eigenvalue λ_2 :

$$y(G, S) = \text{clip}\left(1 - \frac{\lambda_2(G - S)}{\lambda_2(G)}, 0, 1\right).$$

If u_2 is the Fiedler vector, first-order sensitivity gives

$$\hat{y}_{\text{spec}} = \text{clip}\left(\frac{\sum_{(i,j) \in S} w_{ij}(u_{2,i} - u_{2,j})^2}{\lambda_2(G)}, 0, 1\right).$$

The residual model learns

$$\hat{y} = \text{clip}(\hat{y}_{\text{spec}} + \tanh r_\theta, 0, 1).$$

Evaluation design

- ▶ Five seeds: 11, 22, 33, 44, 55.
- ▶ Graph-disjoint synthetic train/validation/test split.
- ▶ Independent and spatial-cluster edge failures.
- ▶ Zero-shot evaluation on five cached OSM networks.
- ▶ Eight learned configurations plus the spectral baseline.
- ▶ Paired differences and 10,000 seed-level bootstrap resamples.

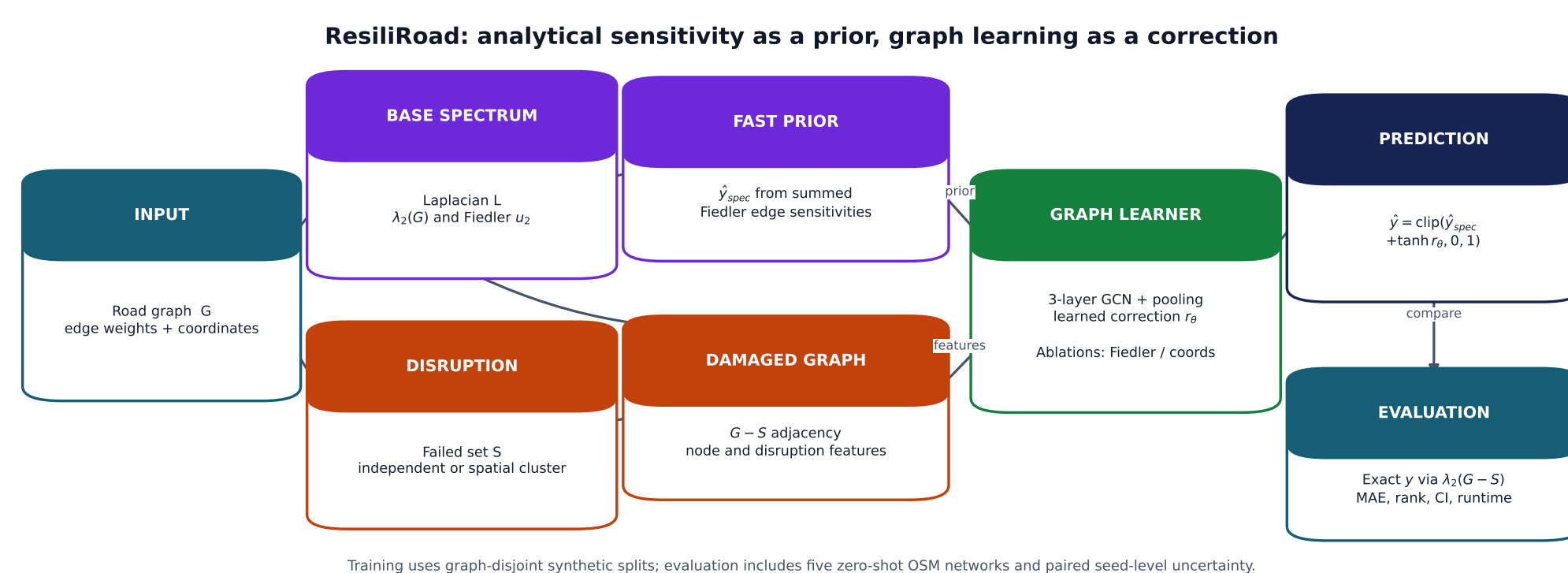
Statistical guardrail

Scenarios sharing a graph or seed are correlated. Confidence intervals resample the five seed-level metrics - not thousands of scenario rows.

Related work and positioning

Algebraic connectivity is a classical global robustness descriptor. Road studies have considered random, targeted, and localized multi-link failures; graph learning supplies nonlinear surrogates. Our contribution is the controlled separation of a Fiedler node feature from an explicit residual prior, evaluated with graph-disjoint splits, multi-area transfer, and seed-paired uncertainty.

Method overview



Baselines and ablations

- ▶ Direct GCN with all node features.
- ▶ Direct GCN without Fiedler information or coordinates.
- ▶ Residual GCN with the same two feature ablations.
- ▶ Deep Sets and graph-summary MLP.
- ▶ Exact eigendecomposition and first-order spectral estimate.

All learned models use the same graph split, optimizer, early stopping, and 80-epoch maximum.

Five real road networks

Site	Nodes	Edges	Density
HCMUS	163	221	.0167
VIASM	206	301	.0143
Da Nang	142	203	.0203
Can Tho	114	178	.0276
Da Lat	48	55	.0488

Each site contributes 100 scenarios per failure process and seed. No OSM sample is used for training, early stopping, or model selection.

Core finding

The robust gain comes from the **explicit residual spectral prior**, not from merely exposing a Fiedler node coordinate. Feature-ablation intervals cross zero, so individual feature necessity is not claimed.

Reproducibility

CPU-only environment and library versions are recorded. Cached GraphML avoids future OSM drift. One script regenerates all five seeds, bootstrap summaries, figures, and the paper. Raw predictions are regenerable and excluded from Git to keep the repository compact.

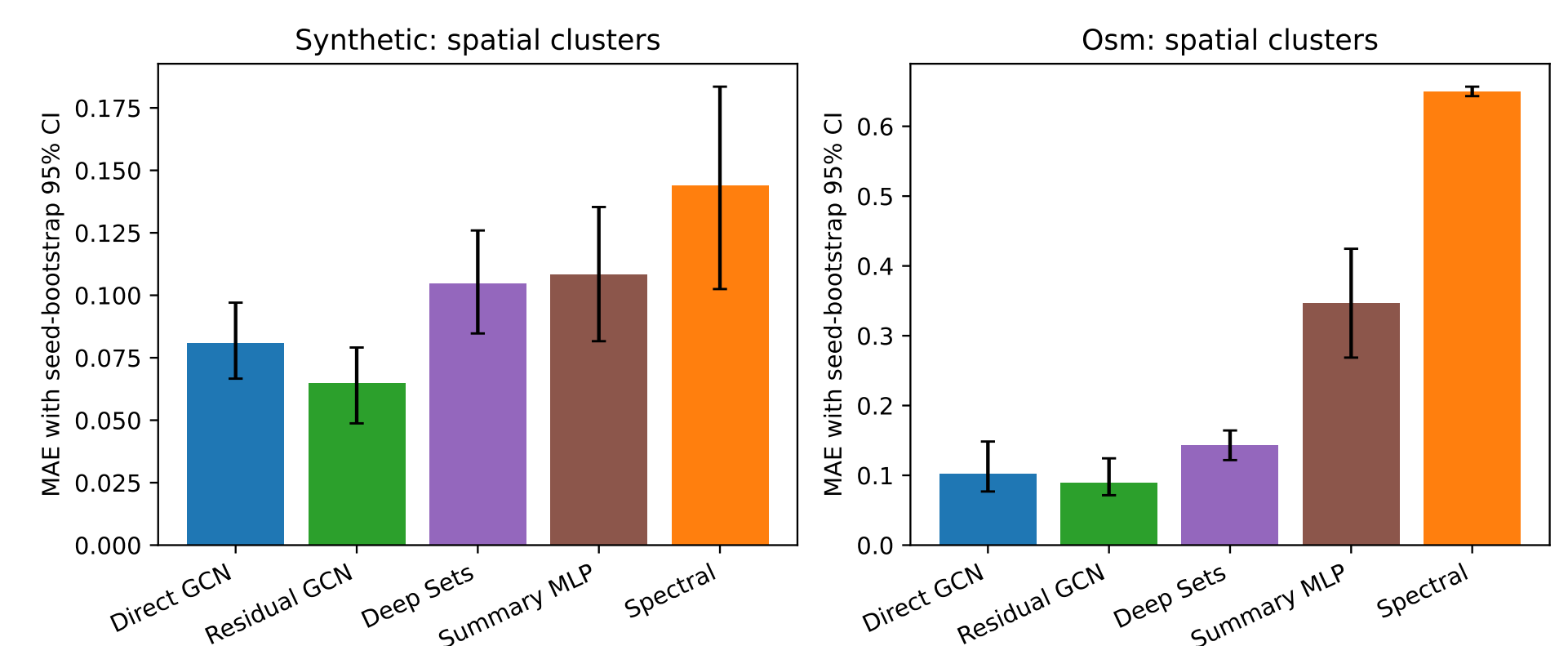
Zero-shot OSM results

Method	Independent Spatial cluster	
	MAE	MAE
Spectral	.502	.650
Summary MLP	.281	.347
Deep Sets	.114	.143
Direct GCN	.106	.102
Residual GCN	.097	.090

Paired residual improvement over direct GCN:

- ▶ Independent: **0.0085** [0.0039, 0.0131].
- ▶ Spatial: **0.0121** [0.0026, 0.0247].

Spatial-disruption comparison



Runtime on CPU

Method	ms/scenario
Exact eigendecomposition	5.094
Spectral first order	0.018
Summary MLP	0.442
Deep Sets	0.430
Direct GCN	0.502
Residual GCN	0.674

Residual inference is about **7.6x faster** than exact recomputation on the evaluated small OSM graphs. No GPU was used.

Where does it fail?

Residual MAE rises from .070 on connected damaged graphs to .112 after disconnection. Da Lat is hardest (.189 MAE), showing that density alone does not explain transfer error. No scenario clipped the spectral estimate at one.

Takeaway, limitations, and code

Residual spectral learning consistently improves direct regression across both failure processes and both domains. The result concerns structural connectivity, not flood risk, traffic demand, capacity, or recovery.

Paper, code, cached OSM, and reproducibility scripts:

github.com/lee-vtruong/
ResiliRoad

